

Commerce XI Maths App

Exercise 1.1

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 1 - Partition Values | Total questions: 18

Question 1 - Q1(i)

For series of observations : 16, 14.9, 11.5, 11.8, 11.1, 14.5, 14, 12, 10.9, 10.7, 10.6, 10.5, 13.5, 13, 12.6

Compute quartile Q_1

Correct answer: 10.9

Explanation and steps

Arrange raw observations when required, locate $i(n+1)/4$, or use $Q_i = L + (h/f)(iN/4 - c.f.)$ for grouped data. The result is 10.9.

Question 2 - Q1(ii)

For series of observations : 16, 14.9, 11.5, 11.8, 11.1, 14.5, 14, 12, 10.9, 10.7, 10.6, 10.5, 13.5, 13, 12.6

Compute quartile Q_2

Correct answer: 12

Explanation and steps

Arrange raw observations when required, locate $i(n+1)/4$, or use $Q_i = L + (h/f)(iN/4 - c.f.)$ for grouped data. The result is 12.

Question 3 - Q1(iii)

For series of observations : 16, 14.9, 11.5, 11.8, 11.1, 14.5, 14, 12, 10.9, 10.7, 10.6, 10.5, 13.5, 13, 12.6

Compute quartile Q_3

Correct answer: 14

Explanation and steps

Arrange raw observations when required, locate $i(n+1)/4$, or use $Q_i = L + (h/f)(iN/4 - c.f.)$ for grouped data. The result is 14.

Question 4 - Q2(i)

The heights (in cm) of 10 students are 148, 171, 158, 151, 154, 159, 152, 163, 171, 145.

Calculate Q_1

Correct answer: 150.25

Explanation and steps

Arrange raw observations when required, locate $i(n+1)/4$, or use $Q_i = L + (h/f)(iN/4 - c.f.)$ for grouped data. The result is 150.25.

Question 5 - Q2(ii)

The heights (in cm) of 10 students are 148, 171, 158, 151, 154, 159, 152, 163, 171, 145.

Calculate Q_3

Correct answer: 165

Explanation and steps

Arrange raw observations when required, locate $i(n+1)/4$, or use $Q_i = L + (h/f)(iN/4 - c.f.)$ for grouped data. The result is 165.

Question 6 - Q3

Monthly consumption of electricity (in units) of families in a certain locality is given below :

205, 201, 190, 188, 194, 172, 210, 225, 215, 232, 260, 230.

Calculate electricity consumption (in units) below which 25% of families lie.

Correct answer: 191

Explanation and steps

Arrange raw observations when required, locate $i(n+1)/4$, or use $Q_i = L + (h/f)(iN/4 - c.f.)$ for grouped data. The result is 191.

Question 7 - Q4

For the following data of daily expenditure of families (in ₹),

compute the expenditure below which 75% of families include their expenditure.

Daily Expenditure (in ₹)	350	450	550	650	750
No. of families	16	19	24	28	13

Correct answer: 650

Explanation and steps

Arrange raw observations when required, locate $i(n+1)/4$, or use $Q_i = L + (h/f)(iN/4 - c.f.)$ for grouped data. The result is 650.

Question 8 - Q5(i)

Calculate quartile Q_1 for the following frequency distribution :

No. of E-transactions per day	0	1	2	3	4	5	6	7
No. of days	10	35	45	95	64	32	10	9

Correct answer: 2

Explanation and steps

Arrange raw observations when required, locate $i(n+1)/4$, or use $Q_i = L + (h/f)(iN/4 - c.f.)$ for grouped data. The result is 2.

Question 9 - Q5(ii)

Calculate quartile Q_2 for the following frequency distribution :

No. of E-transactions per day	0	1	2	3	4	5	6	7
No. of days	10	35	45	95	64	32	10	9

Correct answer: 3

Explanation and steps

Arrange raw observations when required, locate $i(n+1)/4$, or use $Q_i = L + (h/f)(iN/4 - c.f.)$ for grouped data. The result is 3.

Question 10 - Q5(iii)

Calculate quartile Q_3 for the following frequency distribution :

No. of E-transactions per day	0	1	2	3	4	5	6	7
No. of days	10	35	45	95	64	32	10	9

Correct answer: 4

Explanation and steps

Arrange raw observations when required, locate $i(n+1)/4$, or use $Q_i = L + (h/f)(iN/4 - c.f.)$ for grouped data. The result is 4.

Question 11 - Q6

The following is the frequency distribution of heights of 200 male adults in a factory :

Height in cm.	No. of male adults
145–150	4
150–155	6
155–160	25
160–165	57
165–170	64
170–175	30
175–180	8
180–185	6

Find the central height.

Correct answer: 165.625

Explanation and steps

Arrange raw observations when required, locate $i(n+1)/4$, or use $Q_i = L + (h/f)(iN/4 - c.f.)$ for grouped data. The result is 165.625.

Question 12 - Q7(i)

The following is the data of pocket expenditure per week of 50 students in a class.
It is known that the median of the distribution is ₹120.

Expenditure per week (in ₹)	0–50	50–100	100–150	150–200	200–250
No. of students	7	a	15	b	3

Find missing frequency a.

Correct answer: 12

Explanation and steps

Arrange raw observations when required, locate $i(n+1)/4$, or use $Q_i = L + (h/f)(iN/4 - c.f.)$ for grouped data. The result is 12.

Question 13 - Q7(ii)

The following is the data of pocket expenditure per week of 50 students in a class.
It is known that the median of the distribution is ₹120.

Expenditure per week (in ₹)	0-50	50-100	100-150	150-200	200-250
No. of students	7	a	15	b	3

Find missing frequency b.

Correct answer: 13

Explanation and steps

Arrange raw observations when required, locate $i(n+1)/4$, or use $Q_i = L + (h/f)(iN/4 - c.f.)$ for grouped data. The result is 13.

Question 14 - Q8(i)

The following is the distribution of 160 workers according to the wages in a certain factory :

Wages more than (in ₹)	No. of workers
8000	160
9000	155
10000	137
11000	103
12000	57
13000	23
14000	10
15000	1
16000	0

Determine quartile Q_1 .

Correct answer: 10500

Explanation and steps

Arrange raw observations when required, locate $i(n+1)/4$, or use $Q_i = L + (h/f)(iN/4 - c.f.)$ for grouped data. The result is 10500.

Question 15 - Q8(ii)

The following is the distribution of 160 workers according to the wages in a certain factory :

Wages more than (in ₹)	No. of workers
8000	160
9000	155
10000	137
11000	103
12000	57
13000	23
14000	10
15000	1
16000	0

Determine quartile Q_2 .

Correct answer: 11500

Explanation and steps

Arrange raw observations when required, locate $i(n+1)/4$, or use $Q_i = L + (h/f)(iN/4 - c.f.)$ for grouped data. The result is 11500.

Question 16 - Q8(iii)

The following is the distribution of 160 workers according to the wages in a certain factory :

Wages more than (in ₹)	No. of workers
8000	160
9000	155
10000	137
11000	103
12000	57
13000	23
14000	10
15000	1
16000	0

Determine quartile Q_3 .

Correct answer: 12500

Explanation and steps

Arrange raw observations when required, locate $i(n+1)/4$, or use $Q_i = L + (h/f)(iN/4 - c.f.)$ for grouped data. The result is 12500.

Question 17 - Q9

Following is grouped data for duration of fixed deposits of 100 senior citizens from a certain bank :

Fixed deposits (in days)	0-180	180-360	360-540	540-720	720-900
No. of senior citizens	15	20	25	30	10

Calculate the limits of fixed deposits of central 50% senior citizens.

Options

- A. (270, 630)
- B. (270, 360)
- C. (180, 630)
- D. (360, 630)

Correct answer: (270, 630)

Explanation and steps

$N = 100$. Q_1 is the 25th item: $180 + [(25-15)/20] \times 180 = 270$. Q_3 is the 75th item: $540 + [(75-60)/30] \times 180 = 630$. The printed answer 360 in the textbook key is an error.

Question 18 - Q10

Find the missing frequency given that the median of distribution is 1504.

Life in hours	950-1150	1150-1350	1350-1550	1550-1750	1750-1950	1950-2150
No. of bulbs	20	43	100	x	23	13

Correct answer: 81

Explanation and steps

Arrange raw observations when required, locate $i(n+1)/4$, or use $Q_i = L + (h/f)(iN/4 - c.f.)$ for grouped data. The result is 81.